



Audio Recognition using Fully Connected MLP

Audio Cats and Dogs - Kaggle

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Outline

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PROBLEM DESCRIPTION

Cat and Dog Audio Recognition Dataset

Dataset Overview

- **Dataset:** Cat and Dog Audio Recognition (binary classification: **Cat/Dog**).
- **Goal:** Classify audio recordings into cat or dog sounds.
- **Format:** WAV audio files.

Dataset Composition

- **Cat Class:**
 - 164 WAV files
 - Total duration: **1323 seconds**
- **Dog Class:**
 - 113 WAV files
 - Total duration: **598 seconds**

Cat and Dog Audio Recognition Dataset

Audio Processing Challenges

- Variable-length audio recordings.
- Differences in amplitude, background noise, and recording quality.
- Audio signals require transformation (FFT, Spectrogram, Mel, MFCC) for effective feature extraction.

Source: Kaggle Dataset —

<https://www.kaggle.com/datasets/mmoreaux/audio-cats-and-dogs>

Objective of the Project

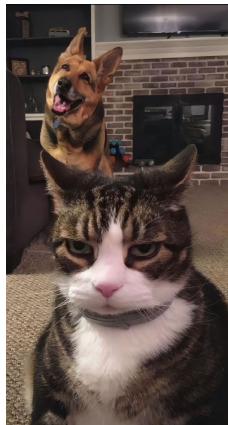
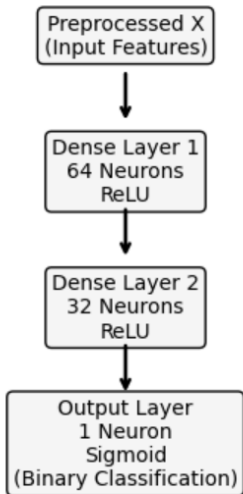
- Compare the performance of a **Fully Connected Neural Network (MLP)** for audio classification using a **Kaggle dataset**.
- Evaluate different **audio representations** (Time Domain, Frequency Domain, Spectrogram, Mel Spectrogram, and MFCC).
- Assess model performance using **Accuracy, F1 Score, and AUC**.

Project Pipeline

- ➊ **Exploratory Data Analysis (EDA)**
- ➋ **Data Preprocessing**
- ➌ **Audio Representation**
 - ➊ Time Domain
 - ➋ Frequency Domain (FFT)
 - ➌ Spectrogram
 - ➍ Mel Spectrogram
 - ➎ MFCC
- ➍ **Model Training**
 - Fully Connected Neural Network ($64 \rightarrow 32 \rightarrow 1$).
- ➎ **Evaluation and Comparison**
 - Metrics: Accuracy, F1 Score, AUC.

MODEL USED

Model



Model

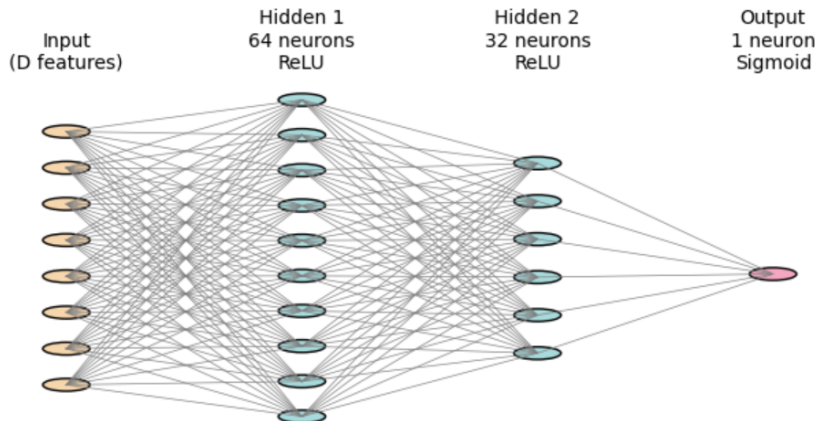


Figure: MLP Architecture for Binary Classification

EDA

Exploratory Data Analysis (EDA)

The analysis of the dataset reveals that audio recordings for both classes (**Cats** and **Dogs**) have highly variable durations. Some samples are very short (around 2 seconds), while others extend beyond 15 seconds. **Multi-Layer Perceptron (MLP)** requires fixed-dimensional input vectors. Neural networks cannot directly process variable-length signals.

Audio Length Distribution from C:\Users\Ale\Documents\Lab01\Dataset_Audio\Mixed\cats

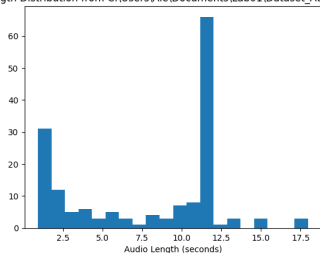


Figure: Dogs original duration distribution

Audio Length Distribution from C:\Users\Ale\Documents\Lab01\Dataset_Audio\Mixed\dogs

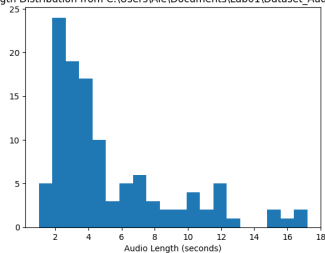


Figure: Cats original duration distribution

DATA PREPROCESSING

Data Preprocessing: Removing Silence

To improve data quality, silence segments were removed from each recording, reducing the effective length of the signals. The total duration of the **Cat** audio files was reduced from **1323.90 seconds** to **909.55 seconds**. Similarly, the total duration of the **Dog** audio files decreased from **598.44 seconds** to **417.95 seconds**.

Although silence removal improves signal quality, the recordings still have variable lengths.

Audio Length Distribution from C:\Users\Ale\Documents\Lab01\Dataset_Audio\no_silence\dogs

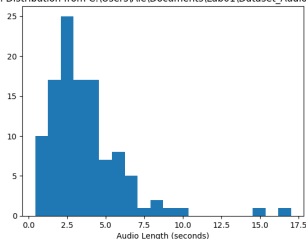


Figure: Dogs distribution without silence

Audio Length Distribution from C:\Users\Ale\Documents\Lab01\Dataset_Audio\no_silence\cats

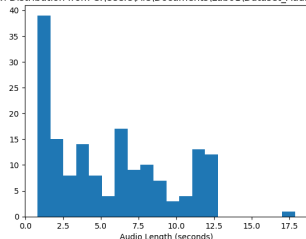


Figure: Cats distribution without silence

Audio Segmentation and Standardization

Segmentation Strategy

- Each audio file is divided into fixed segments of **3 seconds**.
- If the final segment is shorter than 3 seconds, it is **zero-padded**.
- Each segment is saved as an independent WAV file.

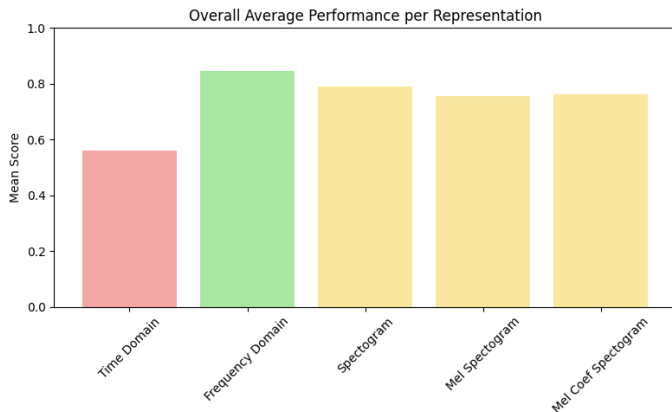
Outcome

- All inputs have uniform duration.
- Enables consistent feature extraction (FFT, Spectrogram, Mel, MFCC).
- Ensures compatibility with MLP training.

PERFORMANCE

Overall Performance

The mean of the evaluation metrics does not fully capture all aspects of model performance. However, it provides a general overview that allows comparison between audio representations by summarizing their overall behavior across Accuracy, F1 Score, and AUC.



1. Time Domain

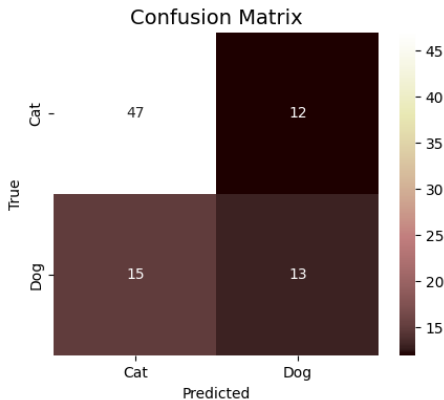


Figure: Confusion Matrix

Model Metrics

Accuracy: 0.690

Precision: 0.520

Recall: 0.464

F1 Score: 0.491

AUC: 0.636

	precision	recall	f1-score	support
0	0.758065	0.796610	0.776860	59.000000
1	0.520000	0.464286	0.490566	28.000000
accuracy	0.689655	0.689655	0.689655	0.689655
macro avg	0.639032	0.630448	0.633713	87.000000
weighted avg	0.681446	0.689655	0.684719	87.000000

2. Frequency Domain

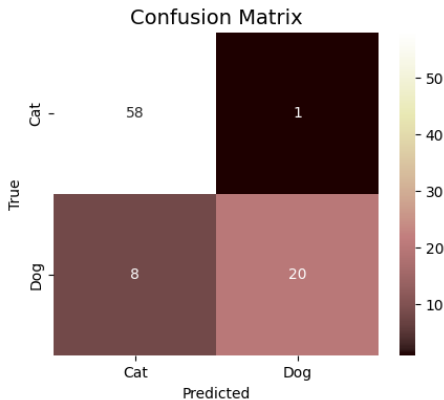


Figure: Confusion Matrix

Model Metrics

Accuracy: 0.897

Precision: 0.952

Recall: 0.714

F1 Score: 0.816

AUC: 0.844

	precision	recall	f1-score	support
0	0.878788	0.983051	0.928000	59.000000
1	0.952381	0.714286	0.816327	28.000000
accuracy	0.896552	0.896552	0.896552	0.896552
macro avg	0.915584	0.848668	0.872163	87.000000
weighted avg	0.902473	0.896552	0.892059	87.000000

3. Spectrograms

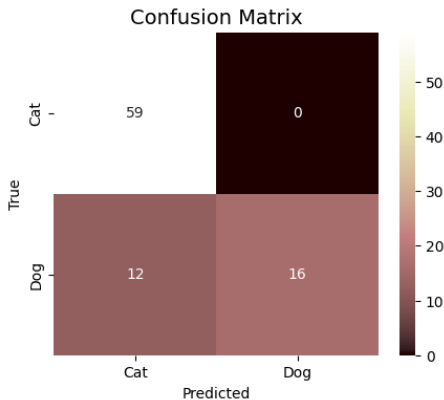


Figure: Confusion Matrix

Model Metrics

Accuracy: 0.862

Precision: 1.000

Recall: 0.571

F1 Score: 0.727

AUC: 0.786

	precision	recall	f1-score	support
0.0	0.830986	1.000000	0.907692	59.000000
1.0	1.000000	0.571429	0.727273	28.000000
accuracy	0.862069	0.862069	0.862069	0.862069
macro avg	0.915493	0.785714	0.817483	87.000000
weighted avg	0.885381	0.862069	0.849626	87.000000

4. Mel Spectrograms

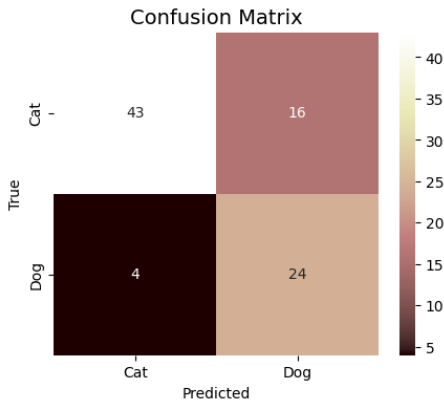


Figure: Confusion Matrix

Model Metrics

Accuracy: 0.770

Precision: 0.600

Recall: 0.857

F1 Score: 0.706

AUC: 0.853

	precision	recall	f1-score	support
0.0	0.914894	0.728814	0.811321	59.000000
1.0	0.600000	0.857143	0.705882	28.000000
accuracy	0.770115	0.770115	0.770115	0.770115
macro avg	0.757447	0.792978	0.758602	87.000000
weighted avg	0.813549	0.770115	0.777387	87.000000

5. MFCC

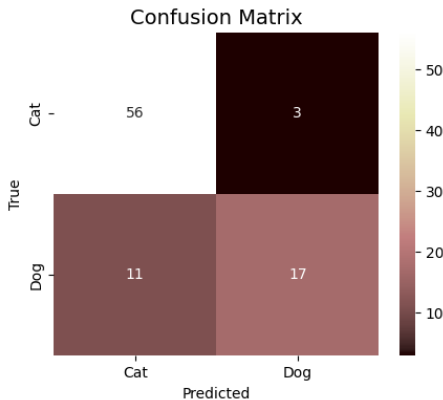


Figure: Confusion Matrix

Model Metrics

Accuracy: 0.839

Precision: 0.850

Recall: 0.607

F1 Score: 0.708

AUC: 0.811

	precision	recall	f1-score	support
0.0	0.835821	0.949153	0.888889	59.000000
1.0	0.850000	0.607143	0.708333	28.000000
accuracy	0.839080	0.839080	0.839080	0.839080
macro avg	0.842910	0.778148	0.798611	87.000000
weighted avg	0.840384	0.839080	0.830779	87.000000

Metrics Comparison

The bar plot shows the individual performance of each audio representation across all evaluation metrics. This visualization allows a detailed comparison of Accuracy, Precision, Recall, F1 Score, and AUC for every feature representation.

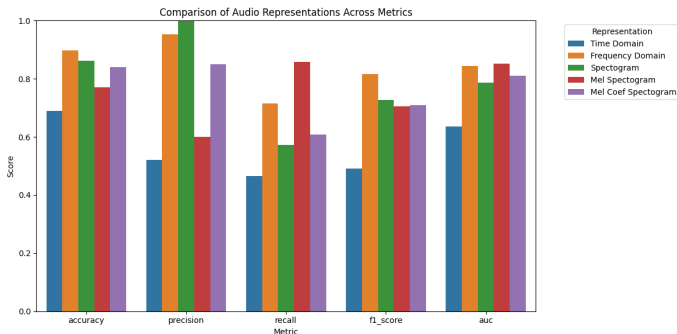
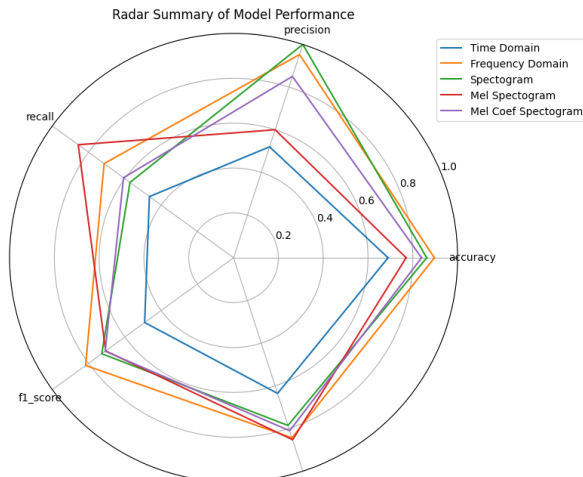


Figure: Performance Comparison Across Metrics and Audio Representations

Metrics Comparison

This representation highlights the balance and stability of each audio representation. Models with a more uniform and larger coverage area indicate more consistent performance across metrics.



ANALYSIS

Comparative Analysis

- The **Time Domain** representation consistently shows the lowest performance across all metrics and is therefore not competitive.
- The **Frequency Domain** representation achieves the highest Accuracy, Precision, and AUC, while maintaining strong F1-score and good Recall.
- The **Spectrogram** representation achieves perfect Precision (1.0), but suffers from lower Recall, indicating a conservative behavior.
- The **Mel Spectrogram** achieves the highest Recall, but with reduced Precision, showing a tendency toward over-detection.
- The **MFCC** representation provides balanced performance but remains slightly below the Frequency Domain model.

Model Selection Strategy

Model selection depends on the cost of classification errors:

- If minimizing **false positives** is critical → Spectrogram (highest Precision).
- If minimizing **false negatives** is critical → Mel Spectrogram (highest Recall).
- If balanced performance is desired across all metrics → Frequency Domain representation.

The Frequency Domain model achieves the best trade-off between Accuracy, F1 Score, and AUC.

Thanks